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National University of Computer and Emerging Sciences

Lahore Campus

Calculus & Analytical Geometry (MT1003)

Final Exam

Date: 31[st] December, 2025

Total Time (Hrs.): 3

Total Marks: 100

Total Questions: 10

Course Instructors

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Mr. Waqar Azeem
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- **Attempt all the questions in the given order**
- **Write question number on your answer with bold faced marker.**

CLO#2: (Explain the core concepts of limits, continuity and use of L'Hopital's Rule to compute some un-determinate forms.)

1.

I. **Give an example of a function $f(x)$ that is continuous for all values of x except one value, where it has removable discontinuity. Explain how you know that f is discontinuous at that value, and how you know the discontinuity is removable.**

II. **Give an example of a function $g(x)$ that is continuous for all values of x except one value, where it has non removable discontinuity. Explain how you know that g is discontinuous there, and how you know the discontinuity is not removable.**

[5+5]

2. Assume that constants a and b are positive. Find equations for all horizontal and vertical asymptotes for the graph of

$$y = \frac{\sqrt{a\,x^2 + 4}}{x - b}$$

[7]

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CLO#3: (Differentiate polynomial, trigonometric, exponential, logarithmic, and inverse functions and to solve problems involving related rates.)

3. A light shines from the top of a pole 50 ft high. A ball is dropped from the same height from a point 30 ft away from the light. How fast is the shadow of the ball moving along the ground 1/2 sec later? (Assume the ball falls a distance $s = 16t^2$ ft in t sec.)

[10]

CLO#4: (Analyze functions using derivatives and asymptotic analysis to determine intervals of increase or decrease, concavity, points of inflection, and to construct accurate graphs.)

4. Graph the rational function using all the steps in the graphing procedure
$$\frac{x^3 - 3x^2 + 3x - 1}{x^2 + x - 2}$$

[10]

5. A window is in the form of a rectangle surmounted by a semicircle. The rectangle is of clear glass, whereas the semicircle is of tinted glass that transmits only half as much light per unit area as clear glass does. The total perimeter is fixed. Find the proportions of the window that will admit the most light. Neglect the thickness of the frame.

[10]

CLO#5: (Riemann sum, evaluation of definite & indefinite integral and their applications to compute lengths of curves / area of regions / volume of solids)

6. For the given function, find a formula for the Riemann sum obtained by dividing the given interval into n equal subintervals. Then take a limit of these sums to calculate the area under the curve over given interval
$$f(x) = x^2 - x^3, \qquad [-1, 0]$$

[8]

7. Find the area of the region in the first quadrant bounded on the left by the $y-axis$, below by the curve $x = 2\sqrt{y}$, above left by the curve $x = (y-1)^2$, and above right by the line $x = 3 - y$.

[10]

8. The region (in the first and second quadrant) bounded by the $x-axis$, $x = y^2$, $x = 3y^2 - 2$ is to be revolved about the $x-axis$ to generate a solid. Which of the methods (disk, washer, shell) could you use to find the volume of the solid? How many integrals would be required in each case? Explain.

[15]

9. Investigate the convergence of the following integral by evaluating

$$\int_0^\infty \frac{dx}{(1+x)\sqrt{x}}$$

[10]

CLO#2: (Explain the core concepts of limits, continuity and use of L'Hopital's Rule to compute some un-determinate forms.)

10. Evaluate the following limits

a)

$$\lim_{x\to 0^+} \frac{x}{e^{-1/x}}$$

b)

$$\lim_{x\to\infty} \left(x - \sqrt{x^2 + x}\right)$$

[10]

Good Luck